D017 — Editorial apparatus

Full returned scope: 51/51 authority pages (printed 55–105). The inherited web-session PASS is not an independent corpus PASS. Every inherited branch and witness remains ZERO_ACCEPTED and is retained unchanged in the original full-state packet. This staged version restores readings checked against the authority image. No claim of correction of the original mathematical paper is made.

Transcription policy

French preserves lexical readings, abbreviations, intrinsic punctuation and formulas, including apparent errors. English translates the assertions and retains the same printed formulas. Display spacing, line wrapping, paragraph indentation and font equivalents are normalized; punctuation separating extracted displays may be reset for sentence continuity. Parenthetical qualifications are preserved. The running Del identifiers and folios are excluded from reader bodies but remain in the complete authority. Roman operator names, blackboard-bold number fields and script mathematical spaces are consistent typographic equivalents. Typewritten statement underlining is represented by italic statement bodies; underlined defined terms are emphasized. Source bibliographic volume underlining is retained and titles remain roman.

Apparent source anomalies retained

Physical 17 has an additional rightward isomorphism arrow ending at no printed node; this is preserved alongside the leftward and downward maps. Physical 18 omits the tilde on SL in the displayed restricted product despite the surrounding discussion of double covers. Physical 29 prints $\gamma' \in$ before a matrix. Physical 32 (theorem 2.4.4) prints $\omega(-1) = k$, in contrast to $(-1)^k$ on physical 30, and prints conductor $n/\ell m < n$ without grouping the slash denominator. Physical 39 gives $0 \leq i < n$ and separately $Ne_{n-1} = 0$. Physical 40 prints unprimed $W(\overline{K}/K)$ in Example 3.1.5 and $W(\overline{K}/K)$ in the proof parenthesis. Physical 42 prints π without subscript u on the right side of (A)(1). Physical 46 omits g inside the left argument of the Hecke induction formula and prints labels 5.2.3.1 and 5.2.6. Physical 49 prints an unprimed dual π in the first denominator. The repeated 3.2.9.8 labels and the unmatched closing-parenthesis glyph in reference [12] are retained. These readings are not silently repaired.

Restoration register

Exact old/new strings, field hashes, and all replayed repairs are supplied in the machine-readable repair logs. The following record states the intervention basis, not an independent final audit verdict.

Physical 1; printed 55. Text-only inherited layer independently reset as editable mathematical TeX from authority; exact replacements are recorded in the accompanying initial six-page typesetting log.

Physical 2; printed 56. Text-only inherited layer independently reset as editable mathematical TeX from authority; exact replacements are recorded in the accompanying initial six-page typesetting log.

Physical 3; printed 57. Text-only inherited layer independently reset as editable mathematical TeX from authority; exact replacements are recorded in the accompanying initial six-page typesetting log.

Physical 4; printed 58. Text-only inherited layer independently reset as editable mathematical TeX from authority; exact replacements are recorded in the accompanying initial six-page typesetting log.

Physical 5; printed 59. A literal superscript comma was accidentally introduced during TeX encoding. Source has exponent n; author confirms intended thinspace command was mistyped. Remove comma, no source emendation. Text-only inherited layer independently reset as editable mathematical TeX from authority; exact replacements are recorded in the accompanying initial six-page typesetting log.

Physical 6; printed 60. A literal superscript comma was accidentally introduced during TeX encoding. Source has exponent n; author confirms intended thinspace command was mistyped. Remove comma, no source emendation. Text-only inherited layer independently reset as editable mathematical TeX from authority; exact replacements are recorded in the accompanying initial six-page typesetting log.

Physical 7; printed 61. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 8; printed 62. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 9; printed 63. Printed operator is Isom, not Iso. Restore parenthetical source mapping rather than comma-separated rewrite. Restore source parenthesis. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 10; printed 64. Preserve underlined defined term as explicit semantic emphasis. Translate source defined-term emphasis. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 11; printed 65. Enlarged native authority crop confirms a tilde above the line, hence source simeq, not equals. Preserve inherited simeq. The initial cold observation was mistaken and is explicitly withdrawn. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 12; printed 66. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 13; printed 67. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 14; printed 68. Source explicitly writes real part of x (despite x having been real); preserve the printed condition. Retain source Re(x) in English too. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 15; printed 69. Source displayed isomorphism is not prefixed by a repeated alpha label. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 16; printed 70. Source group entries and paragraph H use Z/n; preserve its shorthand. Do not change final (Z/nZ)^2 target, which is printed in full. True congruence square has two upward inclusion arrows with hook tails; straight arrows lost visible topology. Preserve four nodes and horizontal maps. Restore source telles que and pour rather than omitting words in display conversion. Restore source condition wording. Preserve underlined defined term as explicit semantic emphasis. Translate source defined-term emphasis. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 17; printed 71. Restore source shorthand Z/n in GL group entries; do not alter printed full Z/nZ in other expressions. Restore source shorthand Z/n in SL group entry. Missing French conditional Si changes a conditional into a declaration. English already preserves If. Restore continuation of the conditional clause. Restore source three-node two-row diagram with left and downward isomorphisms plus the visible rightward arrow ending at no printed node. Do not invent a destination. The exact fallback crop remains authoritative for the stray arrow. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 18; printed 72. Restore source shorthand Z/n in GL group entries; do not alter printed full Z/nZ in other expressions. Restore source parenthetical construction. Restore source word pour and semicolon in 1.2.5.1. Remove supplied second soit. Remove supplied alors. Restore source variable-condition parenthesis. Restore source parenthetical qualifier. Printed restricted product has no tilde on SL; retain this apparent source omission and disclose it instead of silently repairing. Source writes Z/2, not Z/2Z. Restore source index v beneath sum. Restore source index v in summation map. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 19; printed 73. Restore printed telle que in condition. Restore source parenthetical remark. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 20; printed 74. Keep source sum index and separate parenthetical domain in original topology. Preserve underlined defined term as explicit semantic emphasis. Translate source defined-term emphasis. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 21; printed 75. Printed function is uppercase Phi, not lowercase varphi. Preserve underlined defined term as explicit semantic emphasis. Translate source defined-term emphasis. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 22; printed 76. Source une reciproque is indefinite, not a uniquely identified converse. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 23; printed 77. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 24; printed 78. Restore printed pour in both cases of 2.1.7.1. Translate explicit source pour in both cases. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 25; printed 79. Preserve underlined defined term as explicit semantic emphasis. Translate source defined-term emphasis. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 26; printed 80. Preserve underlined defined term as explicit semantic emphasis. Translate source defined-term emphasis. Source and English rechecked against authority; exact field replacements and findings are recorded in the accompanying field-replacement log for physical pages1–26.

Physical 27; printed 81. Preserve authority grammatical form, not a silent correction. Restore printed connective pour outside parentheses. Restore source parenthesis. Restore printed connective pour. Restore source parenthesis around inclusion. Authority last transformation positive i; preserve preceding negative-i direct-sum matrix.

Physical 28; printed 82. Authority support exponent has minus i, not plus i. Preserve printed singular restriction after les. Authority uses bracketed parenthesis and no c'est-à-dire. Restore printed parenthesis and comma. Preserve printed congruence notation rather than ideal-membership rewrite. Restore tilde over the represented Kirillov function. Restore dependent English formula tilde.

Physical 29; printed 83. Restore visible algebraic-closure bar over k. Close authority bracket spanning physical28/29. Restore authority reference punctuation. Printed membership sign before primed matrix; disclose source anomaly. Source parenthetical.

Physical 30; printed 84. Source reads Ker(psi), not K(psi). Preserve abbreviation and parentheses. Preserve parentheses in authority. Restore printed domain parenthesis. Restore printed connective pour. No source words on obtient.

Physical 31; printed 85. DIP-011: abbreviation restoration. Retain authority notation Z(n) rather than silently expanding it. Restore opening source parenthesis. Close source parenthesis. Restore source parenthesis. No lower p on source newform tensor; earlier V tensor retains p. Restore source avec. Defined-term underline mapped to emphasis. Corresponding source term emphasis.

Physical 32; printed 86. Authority theorem2.4.4 visibly reads omega(-1)=k; do not silently substitute mathematically expected (-1)^k. Authority quotient is G_A/GL(2,Q), not the inverted quotient G_Q backslash GL(2,A). Preserve source slash denominator presentation and disclose interpretation. Restore source parenthetical rather than paraphrase. Restore source semicolon and remove où. Restore source parenthetical ending and avec. Source parenthetical. Restore source pour. Source list punctuation.

Physical 33; printed 87. Retain printed grammatical number. Restore original parenthetical wording.

Physical 34; printed 88. Authority right quotient restored, as on page32. Do not silently expand source quotient notation. Unipotent acts on complete real/finite adelic pair, not finite component alone. Restore source parenthetical, remove c est-à-dire addition. Close source Whittaker parenthetical. Restore source parenthetical.

Physical 35; printed 89. DIP-012 abbreviation restoration. Remove added connective absent from authority. Restore original words and parentheses. Restore source avec.

Physical 36; printed 90. Preserve source character O for units. Remove prime absent from printed v_p(x). Preserve printed grammatical number. Source support is O, not Q_p^*. Dependent English support correction from source. Preserve original singular spelling. Restore omitted de V_p. Restore source parenthetical. Close normalization parenthetical. Preserve source grammar.

Physical 37; printed 91. Authority prints f, not f-prime; retain literal source even though f-prime would clarify. Restore source local quotient Z/p^nu_p(N). Restore printed N_p, not NZ_p. Retain printed global quotient notation. Printed quasi-character is omega_k, not psi_k. First source sum has no lower index. Source parenthetical. Source parenthetical instead of added Ici. Close source valuation parenthetical. Remove added alors.

Physical 38; printed 92. DIP-013 straight quotation glyph restoration. No colon in source. Restore source inertia parenthetical. Close source parenthetical. Restore source parenthetical. Source parenthetical condition.

Physical 39; printed 93. DIP-014 restore parenthetical source assertion. Authority uses subscript i, not power of omega_1. Printed range is i<n followed by Ne_(n-1)=0; preserve apparent source inconsistency. Second source direct sum has no lower tau. Prime printed after closing parenthesis in this definition only. Source basis-index parenthetical. Source parenthetical and endpoint n.

Physical 40; printed 94. Restore authority subscript family. Preserve printed grammatical form. Authority reads K in this parenthesis; do not silently correct it to L. Source W-representation hyphen, not ill-typed map into vector space. Translate source group-representation notation, not a group-to-vector-space map. Example prints unprimed W; apparent source anomaly retained. Mirror source unprimed W with apparatus disposition. Source parenthetical. Original source wording; remove added de.

Physical 41; printed 95. Source terminal field letter is F, not K. Restore source soit rather than inserted equality. Preserve source colon and lowercase on a.

Physical 42; printed 96. Authority index in normalization twist is lower-case k. A(1) source right side prints pi without u; no silent completion. DIP-015 source abbreviation and parentheses. DIP-016 source abbreviations and parentheses. Source parenthetical. Source nested parenthetical label.

Physical 43; printed 97. Authority trivial representation uses double absolute-value bars in both factors. Restore source parenthesis; no expansion. Source parenthetical. Source definition has no colon. Source parenthetical, no où. Source unaccented capital and heading punctuation.

Physical 44; printed 98. Source FE dualizes pi_u(sigma) and contains no omega_s tensor factor in numerator. Faithful translation of sans zéro, stronger than not identically zero. Faithful translation of sans zéro. Source parenthetical rather than expanded wording. Source unaccented capital and heading punctuation.

Physical 45; printed 99. No summation subscript is printed. Preserve order of the printed relation.

Physical 46; printed 100. Remove spurious exponent a from the Tate induction character. Hecke induction formula omits g inside the left argument in the printed source; do not silently supply it. Restore printed case label. Translate printed case label. Source parenthetical. Original heading wording, no Cela.

Physical 47; printed 101. Restore Weyl operator w in Hecke functional equation; inherited psi(x) multiplication changes the formula. Remove connective par added in inherited edition. Restore both source abbreviations and parentheses. Restore source parentheses around change of Haar measure.

Physical 48; printed 102. Restore straight quotation marks. Source correspondence uses a single mapsto arrow. Source has unaccented initial A. Physical page49 begins on a; preserve exact boundary. Move one has to corresponding physical page49 boundary. Restore parenthetical source reference. Authority crossed isomorphism sign, not inequality. Source parenthetical.

Physical 49; printed 103. First denominator contains unprimed pi in source; inherited prime is not printed. Preserve source digit 0. Source pairing symbol contains blank arguments, not an inserted comma. Source parenthetical.

Physical 50; printed 104. Preserve printed grammatical number. Restore source parenthesis. Source3.2.9.7 arrow has no mapsto tail. Source repeated injectivity arrow has no mapsto tail. Source parenthetical. Source comma after parenthesis.

Physical 51; printed 105. No final period printed in reference2. Restore reference5 punctuation. Restore reference7 separators. Preserve published title spelling. Restore reference8 separator. Restore reference10 punctuation. No final period printed in reference10. Restore reference12 separator. Printed authority has an unmatched closing-parenthesis glyph here, not supplied volume I; retained with facsimile witness and apparatus disclosure. Preserve source volume underline. Source volume is not emphasized. Do not newly italicize source title. No source comma after year. No source comma after317.

Image fallbacks

These unretouched authority crops preserve fragile alignment, arrows, diagram topology, and source anomalies. They supplement the editable editions; they do not replace reader pages. Crop hashes are replayed against the inherited asset ledger. The complete 51-page authority remains the controlling witness.

Physical 6; printed 60

diagram alignment fallback

$$
\begin{tikzcd}
\mathrm{GL}(n, \mathbb{A})/\mathrm{GL}(n, \mathbb{Q}) & \\
{[\mathrm{GL}(n, \mathbb{R}) \times \mathrm{GL}(n, \hat{\mathbb{Z}})]/\mathrm{GL}(n, \mathbb{Z})} \arrow[u, "\wr"'] \arrow[r] & \mathrm{GL}(n, \mathbb{R})/\mathrm{GL}(n, \mathbb{Z})
\end{tikzcd}
$$

Physical 7; printed 61

quotient isomorphism arrow alignment

Elle induit un isomorphisme

(0.1.4.2) $$GL(n,\hat{\mathbb{Z}})\backslash \mathcal{R}_{\mathbb{A}} \xrightarrow{\sim} \mathcal{R}_{\mathbb{Z}} \ .$$

Physical 12; printed 66

differential operator signs overbars

On pose $\omega_1 = x_1 + iy_1$, $\omega_2 = x_2 + iy_2$. Pour β une forme holomorphe, on note $|\beta|^2$ la forme réelle $\beta \wedge \bar{\beta}$ (et l'élément de volume positif correspondant pour β une 2-forme).

(A_h) $$dg = \left| \frac{d\omega_1 \wedge d\omega_2}{\omega_1 \wedge \omega_2} \right|^2 = 4\, dx_1\, dy_1\, dx_2\, dy_2\, (x_1 y_2 - x_2 y_1)^{-2} .$$

(B_h) $$\det g = x_2 y_1 - x_1 y_2$$

(C_h) $$\lambda.(\omega_1, \omega_2) = (\lambda\, \omega_1, \lambda\, \omega_2)$$

$$(\omega_1, \omega_2)\, \gamma = (a\, \omega_1 + b\, \omega_2,\ c\, \omega_1 + d\, \omega_2)$$

pour $\gamma = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$.

(D_h) $$H * f = [(-\omega_1 \frac{\partial}{\partial \omega_1} + \bar{\omega}_1 \frac{\partial}{\partial \bar{\omega}_1}) + (-\omega_2 \frac{\partial}{\partial \omega_2} + \bar{\omega}_2 \frac{\partial}{\partial \bar{\omega}_2})]\, f$$

$$X * f = [-\, \bar{\omega}_1 \frac{\partial}{\partial \omega_1} - \bar{\omega}_2 \frac{\partial}{\partial \omega_2}]\, f$$

$$Y * f = [-\, \omega_1 \frac{\partial}{\partial \bar{\omega}_1} - \omega_2 \frac{\partial}{\partial \bar{\omega}_2}]\, f$$

1.1.5.2 Demi-plans de Poincaré

Physical 16; printed 70

true level congruence square

vérifie encore qu'il suffit de vérifier (d_4) dans un domai

luisons cette définition en termes plus concrets, pour $K \subset$

rs un entier n et $H \subset GL(2, \mathbb{Z}/n)$ tel que K soit l'ima

par l'application surjective

$$\begin{tikzcd}
GL(2, \hat{\mathbb{Z}}) \arrow[r] & GL(2, \mathbb{Z}/n) \\
K \arrow[u, hook] \arrow[r] & H \arrow[u, hook]
\end{tikzcd}$$

.t K_n défini par $H = \{e\}$. D'après (0.1.4.1), le quotier

Physical 17; printed 71

quotient component alignment

(1.2.4) On notera que l'ensemble de réseaux marqués $K\backslash G_{\mathbb{A}}/GL(2,\mathbb{Q})$ est en général disconnexe.

D'après ce qui précède ou (0.1.4.1), il s'identifie en effet à

(1.2.4.1)
$$\begin{tikzcd} K\backslash G_{\mathbb{A}}/GL(2,\mathbb{Q}) & K\backslash (G \times GL(2,\hat{\mathbb{Z}}))/GL(2,\mathbb{Z}) \arrow[l, "\sim"'] \arrow[r, "\sim"] \arrow[d, "\wr"] & {} \\ & G \times (H\backslash GL(2,\mathbb{Z}/n))/GL(2,\mathbb{Z}) & \end{tikzcd}$$

Les deux composantes connexes de G sont permutées par les éléments de $GL(2,\mathbb{Z})$ de déterminant -1, et $SL(2,\mathbb{Z})$ s'envoye sur $SL(2,\mathbb{Z}/n)$. De (1.2.4.1), on tire donc une bijection

$$\pi_0(K\backslash G_{A}/GL(2,\mathbb{Q})) \xrightarrow[\sim]{\mu} \det(H)\backslash(\mathbb{Z}/n\,\mathbb{Z})^*$$

ou, en termes adéliques

$$\pi_0(K\backslash G_{\mathbb{A}}/GL(2,\mathbb{Q})) \xrightarrow{\sim} \det K\backslash \{\pm 1\} \times (\mathbb{A}^f)^*/\mathbb{Q}^* \ .$$

En terme de réseaux marqués $(R_0,\bar{\alpha})$, cette application se décrit comme

Physical 44; printed 98

tate hecke functional equation layout

prolongement analytique dans la famille de représentations $\pi \otimes \omega_s$, toutes réalisées dans le même espace.

Le quotient

$$\frac{Z(f, \pi_u(\sigma) \otimes \omega_s \otimes \omega_{\frac{1}{2}})}{L(\sigma \otimes \omega_s)}$$

est fonction holomorphe de s , et une combinaison linéaire finie de coefficients de ces endomorphismes, pour f variable, est une fonction sans zéro de s . On a l'équation fonctionnelle

$$\frac{{}^t Z(\hat{f}, (\check{\pi}_u(\sigma) \otimes \omega_1) \otimes \omega_{\frac{1}{2}})}{L(\check{\sigma} \otimes \omega_1)} = \epsilon(\sigma, \psi, dx) \frac{Z(f, \pi_u(\sigma) \otimes \omega_{\frac{1}{2}})}{L(\sigma)}$$

(F) <u>Equation fonctionnelle de Hecke</u>

Soient ψ et dx comme en (E), et σ tel que $\pi_u(\sigma)$ soit de dimension infinie.

Pour $\varphi \in \mathcal{K}(\pi(\sigma) \otimes \omega_{-\frac{1}{2}}) = \mathcal{K}(\pi(\sigma)).\omega_{-\frac{1}{2}}$ (3.2.2.3), le quotient

$$\frac{\int_{K^*} \varphi(x)\ \omega_s(x)\ d^*x}{L(\sigma \otimes \omega_s)}$$

est une fonction holomorphe de s (intégrale définie par prolongement analytique en s). Pour φ convenable, cette fonction de s est sans zéro.

Posons $w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Pour $\varphi \in \mathcal{K}(\pi(\sigma))$, on a l'équation fonctionnelle

$$\frac{\int \omega_1(x)\omega^{-1}(x)\ w\varphi(x).\ \omega_{-\frac{1}{2}}(x).d^*x}{L(\omega_1\ \check{\sigma})} = \epsilon(\sigma, \psi, dx) \frac{\int \varphi(x).\omega_{-\frac{1}{2}}(x)\ d^*x}{L(\sigma)},$$

et donc pour tout caractère χ de F^*

Physical 45; printed 99

character twist and new vector formulae

$$\int \frac{\omega_1(x)\ \omega_\pi^{-1}(x)\ \chi^{-1}(x).\ w\varphi(x).\omega_{-\frac{1}{2}}(x).d^*x}{L(\omega_1\ \chi^{-1}\ \check{\sigma})} = \epsilon(\sigma\ \chi,\psi,dx)\int \frac{\varphi(x)\ \chi(x).\omega_{-\frac{1}{2}}(x)\ d^*x}{L(\sigma\ \chi)}$$

(G) Nouveau vecteur

On suppose K non archimédien et $\pi_u(\sigma)$ de dimension infinie. Le conducteur de $\pi_u(\sigma)$, défini en 2.2.7, coïncide alors avec le conducteur [4] 8.12.1 de σ.

Soit v_o le nouveau vecteur de $\pi_u(\sigma)$. Dans le modèle de Kirillov, $v_o(x)$ est une fonction F'_o de la valuation de x. Son support est contenu dans $\pi^{-n(\psi)}\mathcal{O}$. Posons $F_o(n) = F'_o(n-n(\psi))$, et normalisons v_o pour que $F_o(0) = 1$.
On a

$$\Sigma\ F_o(n)\ q^{-n/2}\ t^{nd} = L(\sigma \otimes \omega^t)\ .$$

Rappelons que $q = p^d$ et que $\omega^t(\pi) = t^d$ $(\omega^{p^{-s}} = \omega_s)$.

(3.2.4) La normalisation choisie de la correspondance $\sigma \longleftrightarrow \pi_u(\sigma)$ a l'avantage de donner lieu à des formules (A) très simples. De (A)(2) résulte aussi que π_u met en correspondance bijective représentations de $PGL(2, K)$ et représentations F-semi-simples de $W'(\overline{K}/K)$ dans $SL(2, \mathbb{C})$.

Physical 46; printed 100

rationalized tate and source numbering

autoduale sur K . On note encore dx la mesure $da\,db\,dc\,dd$ sur $M_2(K)$, et on définit $\hat{f}$ par

$$\hat{f}(x) = \int f(y)\ \psi(\mathrm{Tr}(xy))dy \quad .$$

L'équation fonctionnelle de Tate prend la forme rationalisée

$$(E)_t \qquad \frac{{}^tZ(\hat{f},\ \check{\pi}_t(\sigma) \otimes \omega_2)}{L(\check{\sigma} \otimes \omega_1)} = \varepsilon(\sigma,\ \psi,\ dx) \qquad \frac{Z(f,\ \pi_t(\sigma))}{L(\sigma)}$$

(Pour dx quelconque, multiplier le membre de droite par dx/dx').

(C) (Induction) devient: $\pi_t(\chi_1,\ \chi_2)$ figure dans la représentation de $GL(2,\ K)$, par translation à droite, sur l'espace des fonctions sur $GL(2,\ F)$, K-finies à droite, telles que

$$(5.2.3.1) \qquad f(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} g) = \chi_1\omega_1(a).\chi_2(d).f(g) \quad .$$

(D) (Représentations dégénérées) devient:

a) la représentation unité est $\pi_t(\omega_{-1},\ 1)$.

b) Pour $K = \mathbb{R}$ ou $\mathbb{C}$, et V la représentation évidente de dimension 2,

$$\mathrm{Sym}^n(V) = \pi_t(\omega_{-1}.x^n,\ 1) \qquad \text{(cas réel)}$$

$$\mathrm{Sym}^n(V) \otimes \mathrm{Sym}^m(\bar{V}) = \pi_t(\omega_{-1}.z^n.\bar{z}^m,\ 1) \qquad \text{(cas complexe)}$$

(5.2.6) Définissons la <u>correspondance de Hecke</u> par

$$\pi_h(\sigma) = \pi_u(\sigma) \otimes \omega_{-\frac{1}{2}} = \pi_t(\sigma) \otimes \omega_{-1} \quad .$$

Cette fois, $\pi_h(\chi_1,\ \chi_2)$ figure dans l'espace des fonctions vérifiant

$$f(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}) = \chi_1(a).\chi_2(d)\ \omega_{-1}(d)f(g) \quad ,$$

et, avec les notations de (D) ci-dessus, on a

Physical 47; printed 101

hecke correspondence dfg blocks

$$(D)_h \qquad 1 = \pi_h(1,\ \omega_1)$$

$$\mathrm{Sym}^n\ (\check{V}) = \pi_h(x^n,\ \omega_1) \qquad (K = \mathbb{R})$$

$$\mathrm{Sym}^n(V) \otimes \mathrm{Sym}^m(\bar{V}) = \pi_h(z^n\ \bar{z}^m,\ \omega_1) \qquad (K = \mathbb{C}) \quad .$$

L'équation fonctionnelle de Hecke prend la forme simple suivante. Soit ψ un caractère additif non trivial de K et dx autoduale (pour dx quelconque, multiplier le membre de droite par dx'/dx) . Pour φ dans le modèle de Kirillov de $\pi_h(\sigma)$

$$(F)_h \qquad \frac{\int \omega_1(x)\ \omega_\pi^{-1}(x)\ \chi^{-1}(x)\ w\varphi(x)\ d^*x}{L(\omega_1\ \chi^{-1}\ \check{\sigma})} = \epsilon(\sigma\chi,\ \psi,\ dx) \frac{\int \chi(x)\ \varphi(x)\ d^*x}{L(\chi\sigma)}$$

Avec les notations de (G) , le nouveau vecteur $v_o(x) = F_o(v(x) + n(\psi))$ de $\mathcal{K}(\pi_h(\sigma))$ vérifie

$$(G)_h \qquad \Sigma\ F_o(n)t^{nd} = L(\sigma \otimes \omega^t) \quad .$$

3.2.7 Pour K non archimédien, la correspondance $\sigma \longleftrightarrow \pi_t(\sigma)$ (resp. $\sigma \longleftrightarrow \pi_h(\sigma)$)

Physical 49; printed 103

quaternionic functional equations

on a

$$\frac{{}^t Z(\hat{f},\ \check{\pi}' \otimes \omega_{2-s})}{L(\check{\pi},\ 2-s)} = \varepsilon(s,\ \pi,\ \psi)\ \frac{Z(f,\ \pi' \otimes \omega_s)}{L(\pi,\ s)} \qquad \text{et}$$

$$\frac{{}^t Z(\hat{g},\ \check{\pi} \otimes \omega_{2-s})}{L(\check{\pi},\ 2-s)} = -\ \varepsilon(s,\ \pi,\ \psi)\ \frac{Z(g,\ \pi \otimes \omega_s)}{L(\pi,\ s)}$$

avec des quotients Z/L holomorphes en s et dont une combinaison linéaire est parfois sans 0, comme en (E). ψ désigne un caractère additif de K, et les transformées de Fourier sont prises relativement à une mesure de Haar autoduale pour $\psi\ \mathrm{Tr}(x\ y)$ ou $\psi\ \mathrm{Tr\ red}(x\ y)$.

3.2.9. Voici des références pour les démonstrations. Sauf mention expresse du contraire, on n'exclut pas la caractéristique résiduelle 2.

3.2.9.1. Une démonstration conceptuelle de (3.2.2.1) figure dans [5] Th. 2. La démonstration de loc. cit. vaut pour $GL(n)$.

Physical 51; printed 105

terminal bibliography 9bis to 15

[8] E. Hecke - Mathematische Werke - Vandenhoeck & Ruprecht. Göttingen 1970.

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